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ExplainCheck — Phase 0 Execution, Pilot & Preregistration Package

✅ **Execution status:** All eight immediate actions have been completed to the point possible without an external preregistration account or confirmatory real-dataset runs. The synthetic pilot was executed, metrics were hand-validated, artifacts were packaged, and Protocol v1.0 was locally frozen for Saranraj U's review.

Relationship to the main protocol

This page is the execution companion to 🔬 ExplainCheck — Publication-First Research & Engineering Protocol. The parent page defines the intended study; this page records decisions, pilot evidence, generated artifacts, and the boundary between completed work and work that still requires human approval or external registration.

1. Quantus and OpenXAI review

Quantus

Quantus is a broad quantitative XAI evaluation toolkit. Its paper describes more than 30 metrics organized into faithfulness, robustness, localization, complexity, randomization, and axiomatic categories. It emphasizes that there is no universal metric and that results can depend strongly on parameterization and perturbation baselines.

OpenXAI

OpenXAI is the closest benchmark-level predecessor. It combines a synthetic generator, seven real tabular datasets, logistic-regression and neural-network models, six feature-attribution methods, and metrics for faithfulness, stability, and subgroup disparities. Its synthetic design intentionally uses feature independence to support ground-truth explanations.

ExplainCheck's defensible contribution

ExplainCheck must **not** claim to invent XAI evaluation or tabular XAI benchmarking. The publication contribution is the combination of:

1. Controlled correlation and missingness stress protocols.
2. Prediction-preservation filtering that separates model changes from explanation instability.
3. Individual-feature and grouped-feature correlation analysis.
4. Repeated-seed uncertainty, subgroup effects, and explicit failure reporting.
5. Immutable evidence that automatically generates model cards, LaTeX tables, JSON artifacts, and CI regression decisions.

Sources: Quantus and OpenXAI.

2. Publication strategy decision

Selected strategy: paired artifact strategy.

- **Primary output:** empirical benchmark paper.
- **Supporting artifact:** open-source research software and frozen experiment bundle.
- **Later output:** separate software paper after the API stabilizes and external reuse can be demonstrated.

This keeps the main paper focused on methodological and empirical findings while preserving engineering depth for a software-oriented publication.

3. Dataset selection after license review

Dataset	License	Primary role	Decision
Adult	CC BY 4.0	Mixed types, missing tokens, subgroup analysis	Include
German Credit	CC BY 4.0	Small finance benchmark and subgroup diagnostics	Include

Dataset	License	Primary role	Decision
Bank Marketing	CC BY 4.0	Larger mixed-type and temporal benchmark	Include
Breast Cancer Wisconsin Diagnostic	CC BY 4.0	Natural collinearity stress test	Include for correlation track
Spambase	CC BY 4.0	Dimensionality and runtime sensitivity	Reserve

Dataset-specific safeguards

- Adult is a historical 1994 extract and must not be described as representative of current populations.
- German Credit is small and high-stakes; subgroup results require uncertainty and power checks.
- Bank Marketing's `duration` field is excluded from the primary model because it is known only after contact and can create leakage.
- WDBC is used as a technical correlation benchmark, not as evidence for clinical deployment.
- Spambase includes collection-specific indicators and is unsuitable for broad claims about modern email.

Official UCI sources: Adult, German Credit, Bank Marketing, WDBC, and Spambase.

4. Synthetic linear ground-truth implementation

The executed pilot generates eight independent Gaussian features with true coefficients:

```
[1.50, -1.20, 0.90, -0.70, 0, 0, 0, 0]
```

For each of ten fixed seeds, 3,000 observations are generated and split into 80% training and 20% test data. A custom L2-regularized logistic model is fitted by deterministic full-batch gradient descent.

Two explanation channels are evaluated:

- **Exact linear attribution:** learned coefficient multiplied by deviation from the training mean.
- **Randomized negative control:** within-sample permutation of exact attribution values.

The negative control is a validation device, not a competitive explainer.

5. Hand validation of metrics

Fidelity fixture

```
x        = [2.0, 1.0, 1.0]
weights  = [1.0, 0.5, 0.2]
baseline = [0.0, 0.0, 0.0]
attribution = [2.0, 0.5, 0.2]
```

Masking the top feature changes the logit by 2.0. Masking the top two changes it by 2.5. Therefore:

```
AOPC@2 = (2.0 + 2.5) / 2 = 2.25
```

Computed result: 2.25 — passed.

Stability fixture

After a small prediction-preserving perturbation, the same top two features remain selected:

```
Jaccard@2 = 2 / 2 = 1.0
```

Computed result: 1.0 — passed.

6. Compute and variance pilot

The pilot used ten seeds and evaluation sizes of 50, 100, and 200 explanations per seed.

Model recovery

- Mean ROC AUC: **0.878**.

- The run completed in approximately **2.6 seconds** in the current sandbox.
- This timing is only for the custom linear pilot and must not be extrapolated to KernelSHAP, LIME, or full real-data experiments.

Results at 200 explanations per seed

Method	Deletion fidelity AOPC@3	Top-3 stability	Across-seed SDs
Exact linear attribution	1.764	0.957	0.080 / 0.012
Randomized negative control	0.753	0.259	0.048 / 0.014

The pilot shows the intended directional behavior: the analytic explanation outperforms the randomized control on fidelity and stability. Increasing evaluation size generally reduced across-seed variability. These are **pipeline-validation results**, not confirmatory findings about SHAP or LIME.

7. End-to-end artifact bundle

The package contains:

- Executable pilot source code
- Quantus/OpenXAI comparison
- Dataset license and datasheet review
- Publication-strategy decision
- Protocol v1.0
- Preregistration-ready record
- Raw and tidy results
- Model-performance CSV
- Hand-validation JSON
- Versioned benchmark JSON
- Run manifest and SHA-256 checksums
- Model card
- Publication-ready CSV and LaTeX tables
- Generated Methods and Results snippets

- PDF and PNG pilot figures

ExplainCheck-Phase0-Research-Package.zip

ExplainCheck Phase-0 research package

8. Protocol v1.0 freeze and preregistration boundary

Protocol v1.0 has been locally frozen with fixed questions, datasets, explainers, models, stress levels, seeds, metrics, statistical plan, and exclusion rules. A preregistration-ready record is included in the package.

⚠️ **Not yet externally preregistered:** An AI agent cannot create an accountable research registration on Saranraj U's behalf. Saranraj U must verify the protocol, confirm the citations and licenses, and deposit the package in OSF or another timestamped registry. The resulting DOI or registration URL must then be recorded in the repository and manuscript.

9. Human and AI responsibilities for this phase

Completed by the AI assistant

- Literature comparison and novelty boundary
- Dataset candidate and license review
- Publication-strategy recommendation
- Pilot code and artifact generation
- Metric fixtures and automated checks
- Compute/variance pilot execution
- Protocol and preregistration drafting

Required from Saranraj U

- Read and approve the comparison and protocol
- Verify the full cited papers and dataset documentation

- Review and understand the pilot code
- Approve final thresholds before confirmatory runs
- Create the external preregistration
- Accept responsibility for every manuscript claim and submission

10. Gate before Phase 1

- [ ] Saranraj U approves Protocol v1.0.
- [ ] Quantus and OpenXAI claims are manually verified.
- [ ] Dataset licenses and citations are manually verified.
- [ ] The package is committed to a version-controlled repository.
- [ ] The preregistration is externally timestamped.
- [ ] Pilot artifacts remain separated from confirmatory outputs.
- [ ] Implementation proceeds with one overlapping Quantus/OpenXAI metric for cross-validation.